

OptiCrop: Problem-Solution Fit

Problem

Farmers lack an easy, data-driven way to match crops to their specific soil and climate conditions, leading to suboptimal crop choices.

Solution

A machine-learning classifier trained on soil/climate features that recommends the best-fit crop, delivered through a simple web form so no technical expertise is required from the end user.

Fit Justification

- The dataset (Crop_recommendation.csv) directly maps the 7 inputs a farmer can obtain to a crop label, matching the problem 1:1.
- Logistic Regression achieves 97.27% accuracy, indicating the inputs carry enough signal to solve the stated problem reliably.
- A web form removes the need for the farmer to interact with code or notebooks directly.